

Here there are some Grammars , Non Terminals are enclosed in < > left and right angle brackets.

- You are required to Eliminate left recursion, left factoring and remove ambiguity if available.
- Then Find FIRST() and Follow() .
- Do Recursive Descent Parsing(apply on any Possible Input as per given grammar)
- Also apply LL(1) Parser (apply on any Possible Input as per given grammar)

1.

$E \rightarrow E + T \mid E - T \mid T$

$T \rightarrow T * F \mid T / F \mid F$

$F \rightarrow (E) \mid id \mid id (ArgList)$

2.

$\langle Block \rangle \rightarrow begin \langle StmtList \rangle end \mid begin \langle StmtList \rangle end ;$

$\langle StmtList \rangle \rightarrow \langle Stmt \rangle ; \langle StmtList \rangle \mid \langle Stmt \rangle$

$\langle Stmt \rangle \rightarrow id = \langle Expr \rangle ; \mid id (\langle Args \rangle) ; \mid if \langle Expr \rangle then \langle Stmt \rangle \mid if \langle Expr \rangle then \langle Stmt \rangle else \langle Stmt \rangle$

3.

$\langle Decl \rangle \rightarrow \langle Type \rangle id ; \mid \langle Type \rangle id = \langle Expr \rangle ;$

$\langle Type \rangle \rightarrow int \mid float \mid id$

$\langle Expr \rangle \rightarrow \langle Expr \rangle + \langle Term \rangle \mid \langle Term \rangle$

$\langle Term \rangle \rightarrow id \mid num \mid (\langle Expr \rangle)$

4.

$\langle Control \rangle \rightarrow if \langle Expr \rangle then \langle Control \rangle \mid if \langle Expr \rangle then \langle Control \rangle else \langle Control \rangle \mid$

$\quad while \langle Expr \rangle do \langle Control \rangle \mid repeat \langle Control \rangle until \langle Expr \rangle \mid$

$\quad for id := \langle Expr \rangle to \langle Expr \rangle do \langle Control \rangle$

5.

$\langle Statement \rangle \rightarrow \langle Statement \rangle ; \langle Statement \rangle \mid if \langle Expr \rangle then \langle Statement \rangle \mid id = \langle Expr \rangle$

6.

$\langle Expr \rangle \rightarrow if \langle Expr \rangle then \langle Expr \rangle \mid if \langle Expr \rangle then \langle Expr \rangle else \langle Expr \rangle \mid while \langle Expr \rangle do \langle Expr \rangle$

7.

$\langle Stmt \rangle \rightarrow id = \langle Expr \rangle ; \mid id (\langle ArgList \rangle) ; \mid id [\langle Expr \rangle] = \langle Expr \rangle ;$

8. $\langle Cmd \rangle \rightarrow while \langle Expr \rangle do \langle Cmd \rangle \mid while \langle Expr \rangle do \langle Cmd \rangle end \mid while \langle Expr \rangle do \langle Cmd \rangle else \langle Cmd \rangle$

